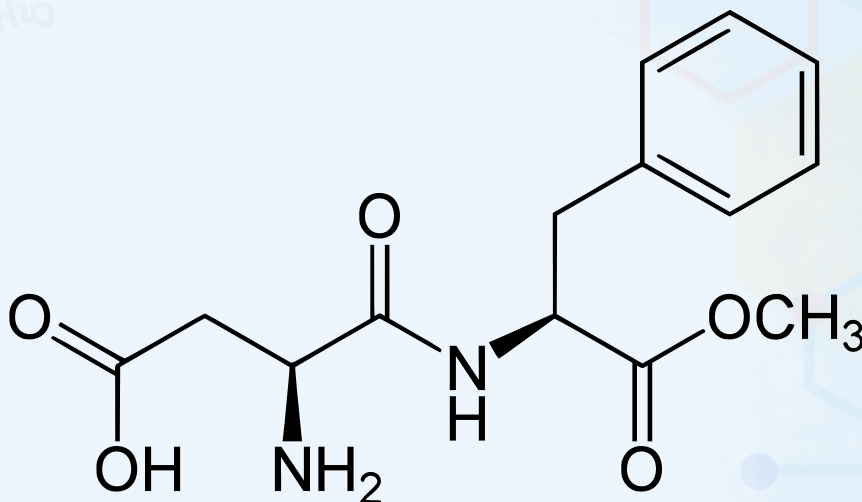


IUPAC name:
methyl L- α -aspartyl-
L-phenylalaninate

Molecular Formula:
C₁₄H₁₈N₂O₅

Molar mass:
294.30 g/mol

Functional groups:
carboxylic acid, ester,
amine, amide, phenyl ring



ASPARTAME

“The Zero-Calorie Dipeptide”

Important Properties

- Slightly sweet organic compound
- **Water-soluble;** more soluble in warm water
- **Melting point:** about 246–247 °C
- Contains polar groups that enable hydrogen bonding
- Low-calorie sweetener

Uses/Value

- Used in sugar-free foods, soft drinks, and chewing gum
- Added to some low-calorie food products
- Valuable commercial sweetener in the food industry
- Provides sweetness with very little energy

Structure & Identity

- **Common name:** Aspartame
- **Family:** Dipeptide methyl ester
- **Appearance:** White crystalline powder
- **Natural sources:** Not naturally occurring; synthetically produced

Risks & Safety

- People with PKU should avoid or restrict aspartame because it contains phenylalanine.
- Safe for the general population when consumed within established limits.
- The FDA-established acceptable daily intake (ADI) is 50 mg/kg body weight/day.
- Aspartame loses sweetness when heated, limiting its use in baked foods.

What Is Amazing About It?

About 200× sweeter than sugar, so only a small amount is needed to achieve the same sweetness. Its structure comes from two amino acids: aspartic acid and phenylalanine

Computer Engineering Relevance

Aspartame production uses embedded systems to monitor temperature, pressure, and other conditions. Computer engineers develop the sensors and control systems for efficient production

References:

- National Center for Biotechnology Information. (2026). PubChem compound summary for CID 134601, Aspartame. <https://pubchem.ncbi.nlm.nih.gov/compound/Aspartame>
- U.S. Food and Drug Administration. (2025, February 27). Aspartame and other sweeteners in food. U.S. Department of Health and Human Services. <https://www.fda.gov/food/food-additives-petitions/aspartame-and-other-sweeteners-food>
- Wikipedia. (2026, March 3). Aspartame. <https://en.wikipedia.org/wiki/Aspartame>